MILITARY SPECIFICATION

INSERT, SCREW THREAD, KEYRING LOCKED, SELF-LOCKING AND NON SELF-LOCKING, 125, 180, AND 220 KSI FTU; GENERAL SPECIFICATION FOR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers locked in, screw thread inserts which have an integral locking keyring to prevent rotation of the insert when installed in the parent material.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-225/6 - Aluminum Alloy Bar, Rod and Wire, Rolled or Cold Finished, 2024.
QQ-A-225/9 - Aluminum Alloy Bar, Rod, Wire and Special Shapes; Rolled, Drawn, or Cold Finished, 7075.
QQ-A-250/6 - Aluminum Alloy, 5083, Plate and Sheet.
QQ-P-35 - Passivation Treatment for Austenitic, Ferritic and Martensitic Corrosion-Resisting Steel (Fastening Devices).
QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-S-764 - Steel, Bar, Corrosion Resisting, Free Machining.
QQ-W-423 - Wire, Steel, Corrosion Resisting.
PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging and Packing for Shipment and Storage of.

Military

MIL-S-5626 - Steel. Chrome Molybdenum (4140) Bars, Rods, and Forging Stock (For Aircraft Applications).
MIL-S-6758 - Steel, Chrome-Molybdenum (4130) Bars and Forging Stock (Aircraft Quality).
MIL-I-6866 - Inspection, Penetrant Method of.
MIL-I-6868 - Inspection Process, Magnetic Particle.
MIL-H-6875 - Heat Treatment of Steels (Aircraft Practice, Process for).

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MIL-A-8625 - Anodic Coatings for Aluminum and Aluminum Alloys
MIL-C-8837 - Coating, Cadmium (Vacuum Deposited)
MIL-S-8879 - Screw Threads, Controlled Radius Root with Increased Minor Diameter, General Specification for.
MIL-L-8937 - Lubricant, Solid Film, Heat Cured.
MIL-N-25027 - Nut, Self-Locking, 250°F, 450°F, and 800°F, 125 KSI FTU, 60 KSI FTU, and 30 KSI FTU.
MIL-W-52263 - Wire, Steel, Corrosion Resisting, Free Machining Stock.

STANDARDS

Federal

Fed. Test Method Std. No. 151 - Metals, Test Methods.

Military

MIL-HDBK-5 - Metallic Materials and Elements for Aerospace Vehicle Structures.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-1312- Fasteners, Test Methods.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

See Supplement 1 for list of associated specification sheets or detail specifications.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards.

Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C 20402).

American National Standards Institute (ANSI) Standard.

ANSI B46.1 - Surface Texture (Surface Roughness, Waviness and Lay)

(Application for copies should be addressed to the American National Standards Institute, Incorporated, 1430 Broadway, New York, New York 10018.)

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Aerospace Material Specifications

AMS2411 - Silver Plating (For High Temperature Applications).
AMS5504 - Steel Sheet, Strip, and Plate, Corrosion and Moderate Heat Resistant.
AMS5610 - Steel, Corrosion and Moderate Heat Resistant.
AMS5737 - Steel Bars, Forgings, and Tubing, Corrosion and Heat Resistant.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., Two Pennsylvania Plaza, New York, New York 10001.)

American Society for Testing and Materials (ASTM) Publications

E10 - Brinell Hardness of Metallic Materials, Test for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Specification Sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets.

3.2 Material.

3.2.1 Inserts Inserts shall be fabricated from materials specified herein.

3.2.1.1 Alloy steel. Unless otherwise specified, inserts shall be made of steel, chrome-molybdenum, composition 4130 conforming with MIL-S-6758 or composition 4140 conforming with MIL-S-5626.

3.2.1.2 Corrosion-resistant steel When specified, MIL-I-45934/1 and MIL-I-45934/2 inserts shall be made of steel, corrosion resistant, composition 303, condition B, conforming with MIL-W-52263 or QQ-S-764

3.2.1.3 Corrosion and heat resistant steel. When specified, inserts shall be made of steel, corrosion and heat resistant, composition A286, conforming with AMS 5737.

3.2.2 Locking keyrings. Locking keyrings shall be made from steel, corrosion resistant, composition 410 conforming with AMS5504, composition 416 conforming with AMS5610, or composition 302 conforming with QQ-W-423.

3.3 Heat treatment. Inserts made of alloy steel in accordance with 3.2.1.1 shall be heat treated in accordance with MIL-H-6875

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3.4 Protective plating or surface treatment. Inserts and locking keyrings shall be furnished with a protective plating or surface treatment as specified herein.

3.4.1 Cadmium plating. Unless otherwise specified, inserts fabricated from alloy steel shall be cadmium plated in accordance with QQ-P-416, Type II, Class 2. The entire insert and keyring shall be plated.

3.4.2 Silver plating. When specified, inserts shall be silver plated in accordance with AMS2411. The entire insert and keyring shall be plated.

3.4.3 Passivation. Inserts fabricated from corrosion-resistant steel and locking keyrings shall be passivated in accordance with QQ-P-35.

3.5 Lubrication. Inserts with self-locking internal threads shall be solid film lubricated. The lubricant shall conform to MIL-L-8937.

3.6 Design, dimensions and tolerances. Inserts and locking keyrings shall be in accordance with the design, dimensions and tolerances specified. These dimensions and tolerances shall apply after application of the protective plating or surface treatment and prior to application of the lubricant as specified in 3.5.

3.7 Construction. The insert, locking keyring and internal thread locking device shall be an integral all-metal unit.

3.8 Threads

3.8.1 Insert internal and external threads shall be right hand.

3.8.2 Internal threads. Internal threads shall conform to Handbook H28, Part I, and MIL-S-8879, except as noted in MIL-N-25027 for self-locking inserts. Sizes and dimensions shall be as specified on MIL-I-45934/1, MIL-I-45934/2 and MIL-I-45934/3. Thread gaging shall be performed after plating and prior to application of the lubricant, as applicable. When a locking feature is employed (see 3.8.7), the GO plug gage shall enter a minimum of 3/4 turn before engagement of the locking device. When the application of a lubricant prevents the use of standard gages, the insert shall permit a minimum free rotational bolt thread engagement of 3/4 turn before engagement of the locking element.

3.8 3 External threads. External threads shall be in accordance with MIL-S-8879 and Handbook H28, Part I.

3.8.4 Thread forming. Threads may be produced either by machining, grinding, or fully formed by a single rolling process prior to plating.

3.8.5 Lead threads. Internal and external lead threads shall not exceed two pitches, and shall not be less than one pitch, including chamfer.

3.8.6 Concentricity. The internal thread pitch diameter shall be concentric with the external pitch diameter within .006 TIR when tested as specified in 4.5.3.

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3.8.7 Thread locking feature. Threads in the area of the locking feature shall be displaced or deformed radially in a manner which is in accordance with the applicable specification sheet. The installed insert shall provide a locking torque with the mating screw or bolt and the locking torque shall not exceed the limits of Table I when tested as specified in 4.6.1. Inserts shall not be subject to locking torque tests after testing as specified in 4.6.2, 4.6.3, or 4.6.4.

TABLE I. Internal Thread Self-Locking Torque
(Inch-Pounds)

Insert Internal Thread	Maximum Locking Torque	Minimum Breakaway Torque
1120-40 UNJC	5	1/
.1380-32 UNJC	10	1 0
.1640-32 UNJC	15	1 5
.1900-32 UNJF	18	2.0
.2500-28 UNJF	30	3 5
.3125-24 UNJF	60	6.5
.3750-24 UNJF	80	9.5
.4375-20 UNJF	100	14.0
.5000-20 UNJF	150	18.0
.5625-18 UNJF	200	24.0
.6250-18 UNJF	300	32.0
.7500-16 UNJF	400	50.0
.8750-14 UNJF	600	70 0
1.0000-12 UNJF	800	90.0

1/ Some indication of torque

3.9 Surface roughness. The surface rougnness of the inserts prior to plating shall not exceed the values stated on MIL-I-45934/1, MIL-I-45934/2, and MIL-I-45934/3, in accordance with ANSI B46 1.

3.10 Mechanical properties Inserts conforming to the design and dimensions of MIL-I-45934/1, MIL-I-45934/2 and MIL-I-45934/3 shall be capable of developing for a load rating, ultimate tensile strength and a minimum proof strength the values for ultimate tensile stress specified herein

MIL-I-45934/1 - 125,000 PSI
MIL-I-45934/2 - 180,000 PSI
MIL-I-45934/3 - 220,000 PSI

3.10.1 Tensile strength (proof test). The installed insert (with or without locking keyring installed) shall develop the tensile strength requirements, as applicable, specified in Table II when tested as specified in 4.6.2.

3.10.2 Resistance to pullout. The installed insert (with or without locking keys installed) shall have a minimum resistance to pullout from the parent material as specified in Table III when tested as specified in 4.6.3.

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TABLE II. Tensile Strength (Proof) Requirements
Insert Internal Thread Proof Test

Insert Internal Thread	Tensile ① Stress Area (A_t), in.^{2}	Tensile Load (Pounds) ②		
		MIL-I-45934/1 Inserts (125 KSI)	MIL-I-45934/2 Inserts (180 KSI)	MIL-I-45934/3 Inserts (220 KSI)
.1120-40 UNJC	.00604		1090	
.1380-32 UNJC	.00909	1140	1640	
.1640-32 UNJC	.0140	1750	2520	
.1900-32 UNJF	.0200	2500	3600	4400
.2500-28 UNJF	.0364	4550	6550	8010
.3125-24 UNJF	.0580	7250	10,400	12,800
.3750-24 UNJF	.0878	11,000	15,800	19,300
.4375-20 UNJF	.1187	14,800	21,400	26,100
.5000-20 UNJF	.1599	20,000	28,800	35,200
.5625-18 UNJF	.203		36,500	44,700
.6250-18 UNJF	.256		46,100	56,300
.7500-16 UNJF	.373		67,100	82,100
.8750-14 UNJF	.509		91,600	112,000
1.0000-12 UNJF	.663		119,000	146,000

① Tensile stress area (A_t) per Handbook H28, Part I, Tables 2.8 and 2.9.

② Tensile Load = (A_t)x(125, or 180 or 220 KSI, as applicable)

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TABLE III. Minimum Resistance to Pullout

Nominal Internal Thread Size	MIL-I-45934/1 Inserts (125KSI)			MIL-I-45934/2 Inserts (180KSI)			MIL-I-45934/3 Inserts (220KSI)		
	External Thread UNJF	Min. Shear Engagement Area-in.^{2} ①	Ultimate Pullout ② Load-lb.	External Thread UNJF	Min. Shear Engagement Area-in^2 ①	Ultimate Pullout ② Load-lb	External Thread UNJF	Min. Shear Engagement Area-in^2 ①	Ultimate Pullout ② Load-lb
.1120-40 UNJC				.1900-32	.0496	1,240			
.1380-32 UNJC	.2160-28	.0583	1,460	.2160-28	.0767	1,920			
.1640-32 UNJC	.2500-28	.0917	2,290	.2500-28	.1144	2,860			
.1900-32 UNJF	.3125-24	.1181	2,950	.3125-24	.1475	3,690	.3125-24	.2187	4,430
.2500-28 UNJF	.3750-24	.2187	5,470	.4375-20	.3030	7,570	.4375-20	.3466	8,660
.3125-24 UNJF	.4375-20	.3030	7,570	.5000-20	.4540	11,300	.5625-18	.5171	12,900
.3750-24 UNJF	.5000-20	.4540	11,300	.6250-18	.6462	16,200	.7500-16	.7850	19,600
.4375-20 UNJF	.6250-18	.6462	16,200	.7500-16	.8629	21,600	.8750-14	1.105	27,600
.5000-20 UNJF	.7500-16	.8629	21,600	.8750-14	1.196	29,900	.8750-14	1.473	36,800
.5625-18 UNJF				.8750-14	1.473	36,800	1.0000-12	1.805	45,100
.6250-18 UNJF				1.1250-12	1.943	48,600	1.1250-12	2.306	57,600
.7500-16 UNJF				1.1250-12	2.716	67,900	1.3750-12	3.299	82,500
.8750-14 UNJF				1.2500-12	3.769	94,200	1.5000-12	4.590	115,000
1.0000-12 UNJF				1.5000-12	4.918	123,000	1.50000-12	5.902	148,000

① Shear engagement area is an assembled dimensional value for the overall engaged area of the mating threaded member. It does not represent a dimension of either of the members in an unassembled condition.

② Pullout load = (Shear engagement area)x(Fsu=25 KSI). To compute minimum pullout load in other materials, multiply shear engagement area by applicable ultimate shear stress (F_{su}) of the material. F_{su} values shall be obtained from MIL-HDBK-5 if available in that document.

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3.10.3 Torque out. The installed insert shall develop the applicable value listed in Table IV as a minimum resistance to torque out in either direction, when tested as specified in 4.6.4.

TABLE IV. Torque Out

Insert Internal Thread Size	Torque (Inch-Pounds)		
	MIL-I-45934/1 Inserts (125 KSI)	MIL-I-45934/2 Inserts (180 KSI)	MIL-I-45934/3 Inserts (220 KSI)
.1120-40 UNJC		20	
.1380-32 UNJC	30	40	
.1640-32 UNJC	45	65	
.1900-32 UNJF	65	90	110
.2500-28 UNJF	125	180	220
.3125-24 UNJF	250	360	440
.3750-24 UNJF	360	540	640
.4375-20 UNJF	600	900	1,000
.5000-20 UNJF	800	1,200	1,200
.5625-18 UNJF		1,600	1,800
.6250-18 UNJF		2,000	2,400
.7500-16 UNJF		3,200	3,600
.8750-14 UNJF		4,800	5,400
1.0000-12 UNJF		6,400	7,200

3.10.4 Hardness. Inserts shall have a hardness range as specified on the applicable Military Specification Sheet when tested as specified in 4.6.5.

3.11 Metallurgical properties.

3.11.1 Discontinuities Inserts shall not contain discontinuities (1) which exceed the following limitations when examined as specified in 4.8.

3.11.1.1 Cracks. Inserts shall be free from cracks in any direction or location.

3.11.1.2 Laps, seams and inclusions. Inserts shall not have laps, seams and inclusions in locations as shown in Figure 1. Laps, seams and inclusions in other locations shall not exceed 20% of the thread height (sharp V thread).

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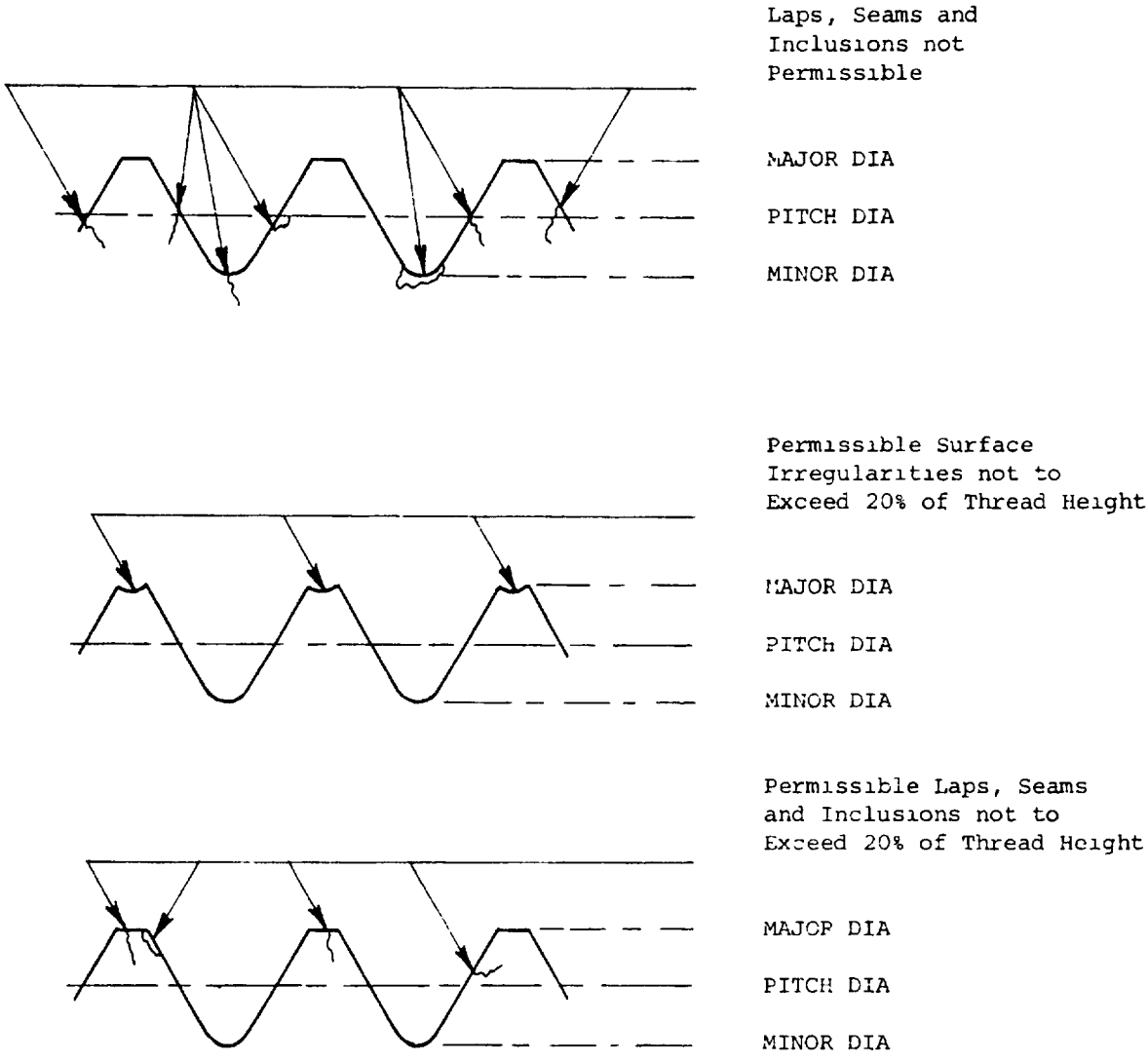


FIGURE 1. Laps, Seams, Inclusions and Surface Irregularities in Thread

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3.12 Workmanship. Workmanship shall be consistent with the type of product, finish, and class of thread fit specified. Inserts shall be of uniform quality and free from defects which would be detrimental to the performance of the inserts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection terms. The definition of terms used in this section shall be in accordance with MIL-STD-109.

4.3 Acceptance tests. Test methods for acceptance shall consist of examination and tests.

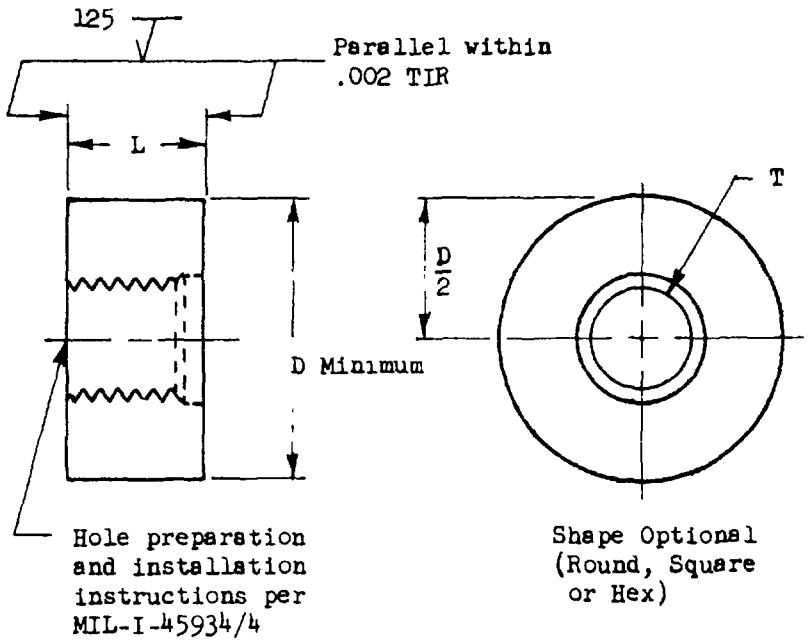
4.4 Sampling.

4.4.1 Lot. A lot shall consist of finished inserts which are of the same material, type, size, fabricated by the same process, heat treated in the same manner, and produced as one continuous run or order, or part thereof, and submitted for acceptance tests at the same time.

4.4.2 Sampling for test materials.

4.4.2.1 Test blocks. Test blocks shall be fabricated as specified in Figures 2 and 3, as applicable. Larger test blocks for multiple testing of inserts are permissible except for test of 4 6.2 and 4.6.3.

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1. Dimensions.

T - Nominal thread diameter of applicable insert external thread.
D - 4 x T (For T thread sizes smaller than .500 dia.).
3 x T (For T thread sizes .500 dia. and larger)
L - Length of applicable insert plus .063.

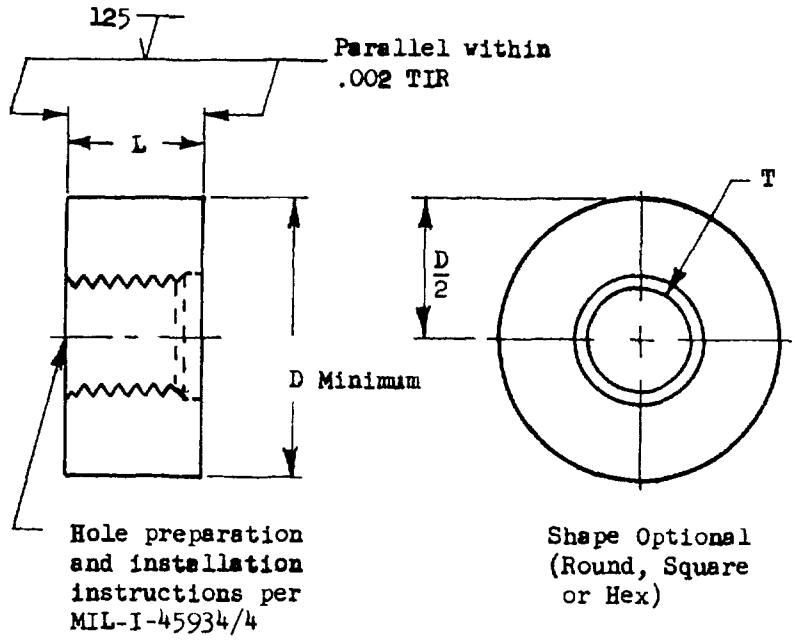
2. Material:

Axial (proof) test (3.10 1) - aluminum alloy, 2024, per QQ-A-225/6 or 7075 per QQ-A-225/9.
Pullout test (3 10.2) - aluminum alloy, 5083, per QQ-A-250/6.

3. Anodize per MIL-A-8625, Type I, Class 1.

FIGURE 2. Axial Load Test Block

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1. Dimensions:

T - Nominal thread diameter of applicable insert external thread.
D - 4 x T (For T thread sizes smaller than .500 dia.).
3 x T (For T thread sizes .500 dia. and larger).
L - Length of applicable insert minus .063 for rotational resistance tests (3.10.3).
L - Length of applicable insert plus .063 for internal locking and breakaway torque tests (3.8.7).

2. Material:

Aluminum alloy, 5083, per QQ-A-250/6.

3. Anodize per MIL-A-8625, Type I, Class 1

FIGURE 3. Torque Test Blocks

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4.4.2.2 Test specimens. Test specimens of inserts taken in accordance with 4.4.4 shall be installed in conformance with MIL-I-45934/4 in test blocks specified in 4.4.2.1.

4.4.2.3 Test bolts and screws. Bolts or screws for use in all tests shall have Class 3A threads.

4.4.2.3.1 Bolts and screws used for tensile testing as specified in 4.6.2, 4.6.3 and 4.6.4 shall be heat treated to 260,000 PSI minimum tensile strength and shall be cadmium plated per MIL-C-8837, Type I, Class 2.

4.4.2.3.2 Bolts or screws used for tests as specified in 4.6.1 shall be of 125,000 PSI minimum tensile strength and shall be cadmium plated per QQ-P-416, Type II, Class 2.

4.4.3 Sampling for examination Sampling for examination of inserts shall be in accordance with MIL-STD-105 at Inspection Level II.

4.4.4 Sampling for test. Samples of inserts shall be taken from a single production lot for testing in accordance with Level II of MIL-STD-105. The Acceptable Quality Level (AQL) shall be 1.5 percent defective

4.4.5 Sampling for packaging and packing. Sampling for packaging and packing shall be in accordance with PPP-H-1581.

4.5 Examination.

4.5.1 External threads External threads shall be checked for thread form per MIL-S-8879 and Handbook H28, Screw-Thread Standards for Federal Services.

4.5.2 Internal threads. Internal threads shall be checked in accordance with Handbook H28 where applicable with minor diameters as specified in MIL-S-8879, except as noted in MIL-N-25027 for self locking inserts.

4.5.3 Concentricity. The concentricity of the internal and external thread pitch diameters shall be checked following examination per 4.5.1 and 4.5.2. The internal thread pitch diameter shall be inspected for concentricity with the external thread pitch diameter using a comparator and suitable thread form chart. Concentricity shall be checked while rotating the insert on a fixed internal plug gage and observing the total eccentricity of the external thread pitch line relative to the thread form chart.

4.5.4 Finish, dimensions, and surface roughness. Finish, dimensions, and surface roughness shall be checked visually and by means of applicable gages.

4.5.5 Plating

4.5.5.1 Cadmium plating. The cadmium plating shall be checked for conformance with the requirements of 3.4.1.

4.5.5.2 Silver plating. The silver plating shall be checked for conformance with the requirements of 3.4.2.

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4.5.6 Lubrication. The solid film lubricant coating shall be examined visually and microscopically at a magnification of 10X for uniformity in color, smoothness, and thickness, and evidence of cracks, bubbles, blisters, runs, foreign matter, grit, separation of ingredients, and any other surface imperfections.

4.5.7 Classification of defects. The classification of defects and the Acceptable Quality Level (AQL) for locked in inserts shall be in accordance with Table V. Any insert containing one or more defects shall be considered a defective unit. The number of defective units which will reject a lot shall be in accordance with Tables I and IV-A of MIL-STD-105.

TABLE V. Classification of Defects

Categories	Defects	Method of Inspection
Critical	None Defined	
Major	AQL = 1.5 percent	
101	Surface finish, plating (3.4)	Visual
102	Lubrication (3.5)	Visual
103	Thread size and form (3.8)	Measure
104	Locking element present (3.8.7)	Visual
105	Surface roughness (3.9)	Measure or Comparison
106	Imperfect threads (3.11.1)	Measure
107	Locking keyrings missing (3.12)	Visual
Minor	AQL = 4.0 percent 1/	
201	Overall length (3.6)	Measure
202	Chamfer on thread ends (3.8.5)	Visual
203	Burrs and tool marks (3.12)	Visual

1/ 4.0 percent AQL shall apply to the total number of defective units, including major defects, which will reject the lot.

4.5.8 Examinations and tests for preparation for delivery. The inspection and testing of the preservation, packaging, packing, and marking shall be in accordance with PPP-H-1581.

4.6 Test methods.

4.6.1 Internal thread self-locking test. Test specimens installed into test blocks (Figure 3) as specified in 4.4.2.2 shall be used for torque testing of the internal locking feature when applicable.

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4.6.1.1 Screw-locking torque. The locking torque test in 3 8.7 shall consist of a 15 cycle room temperature torque test, using test bolts or screws conforming to 4.4.2.3 with sufficient thread length to extend beyond the locking feature a minimum of two (2) pitches (including bolt thread chamfer). A new bolt or screw and a new test specimen shall be used for each complete 15 cycle test. Bolts and screws shall assemble freely, with the fingers, up to the locking feature. The bolt or screw shall be engaged or disengaged from the assembled insert self-locking area for 15 full installation and removal cycles with no axial load on the insert. The test shall be run at a rate slow enough to yield a dependable measure of torque and avoid heating of the bolt. A bolt shall be considered fully installed when two threads extend past the end of the locking feature of the insert, the removal cycle shall be considered complete when the locking feature is disengaged. The maximum locking torque encountered on any installation or removal cycle shall not exceed the values specified in Table I. The minimum breakaway torque for any cycle shall not be less than the values specified in Table I.

4.6.2 Axial strength (proof) test. Test specimens installed into test blocks (Figure 2) as specified in 4.4.2.2 shall be used for axial strength testing. Test shall be in accordance with Test No 8, MIL-STD-1312. The test bolt shall assemble freely into the insert prior to engaging the self-locking device with finger torque and shall be of sufficient length to extend through the insert a minimum of two (2) pitches (including bolt thread chamfer) The bushing and test block clearance holes in the upper and lower yokes (see Figure 4) shall have a free fit not in excess of 0.060 inch greater than the bushing and testblock diameters The bolt clearance hole in the upper bushing (see Figure 4) shall be .006 to .015 inch diameter larger than the nominal diameter of the test bolt. The minimum axial load specified in Table II, as applicable, shall then be applied to the assembly and failure of the insert shall not occur below those loads specified in Table II In the event of bolt failure below the applicable axial strength rating, the test shall be repeated until the applicable axial strength rating of the insert is exceeded. Rate of loading shall not exceed 100,000 PSI per minute per square inch based on the shank area of the bolt.

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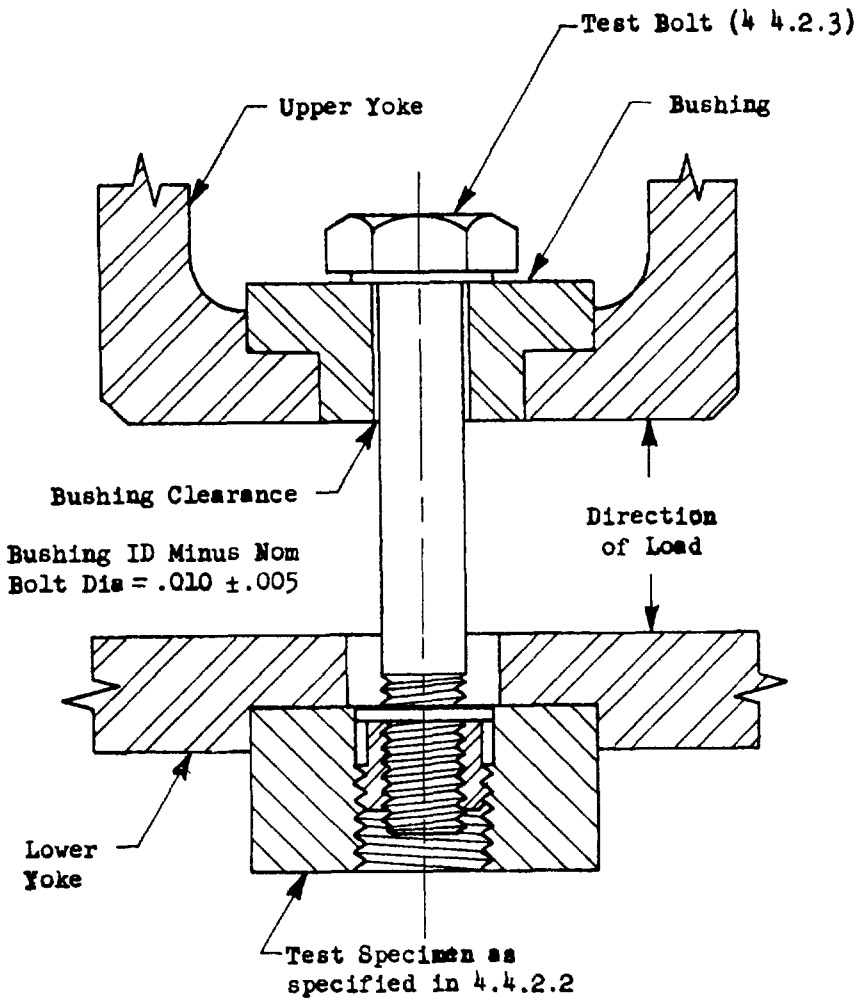
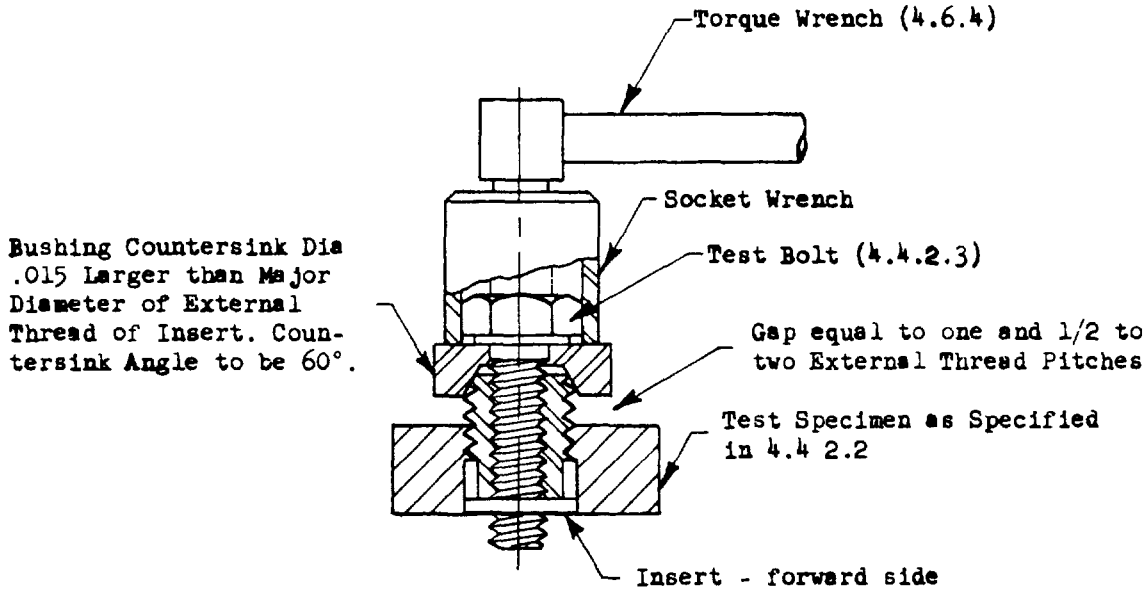


FIGURE 4. Axial Load (4.6.2) and Resistance to Pull-Out Test Fixture (4.6.3)

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4.6.3 Resistance to pullout test. Test specimens installed into test blocks (Figure 2) as specified in 4.4.2.2 shall be used for resistance to pullout tests. The procedure used shall be the same as required for 4.6.2 except the minimum resistance to pullout strength specified in Table III, as applicable, shall be applied to the assembly. Failure of the insert to stay in the test block shall not occur below the loads specified in Table III. In the event of bolt failure below the applicable pullout strength rating, the test shall be repeated until the minimum pullout strength rating of the insert is exceeded. Rate of loading shall not exceed 100,000 PSI per minute per square inch based on the shank area of the bolt. The 5083 aluminum alloy stock used in the test blocks shall be tested to determine its ultimate shear strength and test values shall be adjusted accordingly to correspond with a value for ultimate shear strength of 25,000 PSI.

4.6.4 Rotational resistance test. Test specimens installed into test blocks (Figure 3) shall be used for rotational resistance tests. Torque-out values, with no axial load on the insert, shall not be less than the values specified in Table IV using a test assembly as illustrated in Figure 5. When the near side face of the insert is installed, the far side of the insert shall extend a distance of one and one-half to two external threads of the insert past the far side of the test block allowing it to fit into the countersink of the bushing. The insert shall fit into the bushing countersink to prevent it from expanding under pressure. The test bolt shall enter the insert from the back side (opposite normal entry), and shall be torqued in a clockwise direction. Failures at values below those specified in Table IV shall be cause for rejection.



NOTE: Above test procedure used with insert in the as received condition.

FIGURE 5. Torque Test Fixture (4.6.4)

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4.6.5 Hardness test. Samples of inserts taken in accordance with 4.4.4 shall be tested for hardness requirements of 3.9.4. The test procedure shall be Test No. 6 of MIL-STD-1312 for alloy steel and ASTM E10 for corrosion-resistant steel.

4.7 Chemical analysis. The sample insert taken in accordance with 4.4.4 shall be tested for the composition requirement of 3.2.1 and 3.2.2. The test procedure shall be by Method 111.2 or 112.2 of Federal Test Method No. 151.

4.8 Discontinuities. Magnetic particle inspection performed in accordance with MIL-I-6868 for alloy steel and penetrant particle inspection performed in accordance with MIL-I-6866 for corrosion-resistant steel shall be used to determine the presence of cracks and discontinuities such as laps, seams and inclusions.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Unless otherwise specified, preservation, packaging, packing and marking shall be in accordance with PPP-H-1581.

5.2 Levels. Levels of packaging and packing shall be as specified by the procuring activity. (See 6.2)

6. NOTES

6.1 Intended use. Inserts covered by this specification are intended as a general purpose fastener with mechanical key ring locked feature to resist rotation.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of the specification.
b. Title and number of applicable Military Specification Sheet.
c. Insert material, if other than alloy steel (See 3.2).
d. Quantity required.
e. Selection of applicable levels of preservation, packaging and packing required (See 5.1).

6.3 Definitions.

6.3.1 Crack. A crack is defined as a clean crystalline break passing through the grain or grain boundary without the inclusion of foreign elements.

6.3.2 Lap. A lap is a surface defect appearing as a seam, caused by the folding over of metal fins or sharp corners and then rolling or forging them into the surface, but not welding them.

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6.3.3 Seam. A seam is an unwelded fold or lap which appears as an opening in the raw material as received from the source.

Custodians:

Army - - WC

Navy - AS

Air Force - 82

Review Activities:

Army - AT, AV, EL, MU

Navy - None

Air Force - 11

DSA-IS

NSA

Preparing activity:

Army - WC

Project No. 5340-0866

User activities

Army - GL, ME

Navy - MC, SH

Air Force - None

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 61 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE C-8279

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